

WENXIANG TAO

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PROFILE

AI undergraduate interested in human-AI interaction, interactive intelligent systems, and multi-agent systems. Research experience spans ambiguity-aware human-AI interaction and interactive representations for shared-memory reflection; engineering experience includes campus-scale agent-to-agent infrastructure and student-facing products serving 1,300+ and 600+ users.

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) **Guangzhou, China**
B.Eng. in Artificial Intelligence, GPA: 3.86/4.3 *Expected Summer 2028*
• **Coursework:** Machine Learning, Deep Learning, Human-Centered AI, Statistics and Research Methods, Data Structures and Algorithms

RESEARCH EXPERIENCE

Undergraduate Research Intern **Mar 2026 – Present**
APEX AI-Support-Learning/Education Group, HKUST(GZ) *Prof. Mingming Fan*
• Conduct HCI and human-AI research on intention alignment, ambiguity resolution, and collaborative intelligent interfaces; contribute to research framing, prototyping, and evaluation design.
• Designed and prototyped a Chinese-knot-inspired interactive visual representation that externalizes divergent and shared memories into a jointly editable form for collaborative reflection and co-construction.

Undergraduate Research Trainee, X Program **Jan 2026 – Jun 2026**
Data Science and Analytics Thrust, HKUST(GZ)
• Built a taxonomy-guided two-agent pipeline for interactive Text-to-SQL ambiguity resolution; investigated candidate worlds, ambiguity taxonomies, and expected information gain to identify executable semantic ambiguities and reduce unnecessary clarification.

Undergraduate Research Trainee, X Program **Jun 2025 – Dec 2025**
Artificial Intelligence Thrust, HKUST(GZ)
• Built AccentFlow, a resumable corpus-curation pipeline combining YouTube Data API discovery, text-grounded LLM weak supervision, and deterministic evidence-based filtering. Engineered an offline speech-processing stack that produced 1,000+ hours of sentence-level audio clips.

SELECTED RESEARCH & SYSTEMS

Interactive Visualization for Collaborative Shared-Memory Reflection **2026**
Work in progress
• Designed and prototyped an interactive visual representation that maps divergent and shared recollections onto Chinese-knot-inspired strands, enabling users to inspect, negotiate, and co-construct representations of shared memories.

Campus A2A Assistant Network **2026**
Campus-scale multi-agent system
• Built and deployed a campus assistant network used by **600+ users**; designed personal-agent routing, inter-agent communication, and centralized backend orchestration for accessing campus knowledge and services.

UniKorn TechG Hub **2025**
Live site
• Built and operate a student learning and life community used by **1,300+ users**, shipping academic experiences exchanging, courses planning, Wiki, events/team-up, and campus information features.

KBRD: KB-Grounded Readable De-identification **2026**
Code
• Developed and evaluated a two-stage LLM planner-writer pipeline with knowledge-base grounding and verification on 1,000 feedback records, reducing leakage from **11.1%** to **7.2%** versus the strongest baseline.

TECHNICAL SKILLS

Programming: Python, C++, TypeScript/JavaScript, SQL **ML/AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face, RAG, LLM/agent workflows **Development:** React/Next.js, FastAPI, Git/GitHub, LaTeX

LEADERSHIP & HONORS

• **Founding President**, Network of Developers & Engineers (NODE), HKUST(GZ); founded a student developer community for collaborative campus product building.
• Honorable Mention, COMAP MCM/ICM, 2026; Dean's List, HKUST(GZ), 2024–25 and 2025–26; First Prize, NOIP Senior Division, 2019 and 2021.